

YULEI QIN (秦玉磊)

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RESEARCH INTERESTS

I am now focusing on building LLM-driven agent foundation models for solving long-range, real-world tasks. My previous research scope includes learning with noisy labels and medical image analysis.

EDUCATION

Shanghai Jiao Tong University, Ph.D. in Control Science and Engineering 09/2016 - 03/2022

Thesis: *Chest CT Image Segmentation and its Applications*

With Prof. Jie YANG, Institute of Image Processing and Pattern Recognition

Université de Lyon, INSA Lyon, Ph.D. in Traitement du Signal et de l'Image 11/2019 - 08/2021

Thesis: *Investigation of Deep Learning Methods for Classification and Segmentation of Chromosome and Pulmonary Images*

With Prof. Yue-Min ZHU, CREATIS

Southeast University, B.Eng. in Automation 09/2012 - 06/2016

Thesis: *Research on CT Image-based Sparsifying Transform Learning Methods*

With Prof. Siyu XIA, School of Automation

PROFESSIONAL EMPLOYMENT

Tencent, Youtu Lab, Senior Researcher 04/2022 - now

Development of LLM-driven Agent Models and Systems: *Youtu-Agent RL, Coding Agent, Cloud ADP*

Post-training Research on LLMs: *Complex Instruction Following, Data Selection, Knowledge Distillation*

Multi-modal Data Governance: *MLLM Benchmarking, Noisy Web Learning*

AWARDS

Gold Reviewer, ICML 2026

Outstanding Reviewer, IEEE Transactions on Medical Imaging 2022,2023,2024

Texpert Project, Tencent 2022

Outstanding Graduate, Shanghai Jiao Tong University 2022

Lee Chieh-Ku Thesis Award, Shanghai Jiao Tong University 2022

Outstanding Reviewer, MICCAI 2022

National Scholarship, Ministry of Education 2019

Hella Scholarship, Hella GmbH & Co. KGaA (Shanghai) 2015

PUBLICATION SUMMARY

I have (first/co)-authored 20+ journal articles and conference papers. A detailed list is available on my **Google scholar**. Furthermore, I am an inventor on 18 Chinese patents in various stages of the process.

SELECTED JOURNAL PAPERS

Qin, Y., Yang, Y., Guo, P., Li, G., Shao, H., Shi, Y., ... and Sun, X., Unleashing the Power of Data Tsunami: A Comprehensive Survey on Data Assessment and Selection for Instruction Tuning of Language Models, *Transactions on Machine Learning Research*. 2024

Cui, X., Qin, Y.*, Gao, Y., Zhang, E., Xu, Z., Wu, T., ... and Li, H., SinKD: Sinkhorn Distance Minimization for Knowledge Distillation, *IEEE Transactions on Neural Networks and Learning Systems*. 2024

Qin, Y., Zheng, H., Gu, Y., Huang, X., Yang, J., Wang, L., Yao, F., Zhu, Y.M. and Yang, G.Z., Learning Tubule-Sensitive CNNs for Pulmonary Airway and Artery Vein Segmentation in CT, *IEEE Transactions on Medical Imaging*. 2021

Qin, Y., Wen, J., Zheng, H., Huang, X., Yang, J., Wu, L., Song, N., Zhu, Y.M., Yang, G.Z., Varifocal-Net: A Chromosome Classification Approach using Deep Convolutional Networks, *IEEE Transactions on Medical Imaging*. 2019

Qin, Y., Zheng, H., Huang, X., Yang, J., Zhu, Y.M., Pulmonary Nodule Segmentation with CT Sample Synthesis using Adversarial Networks, *Medical Physics*. 2018

SELECTED CONFERENCE PAPERS

- Qin, Y., Tan, X., He, Z., Li, G., Lin, H., Li, Z., ... and Sun, X. (2025). Learn the ropes, then trust the wins: Self-imitation with Progressive Exploration for Agentic Reinforcement Learning, *International Conference on Learning Representations (ICLR)*. 2026
- Qin, Y., Li, G., Li, Z., Xu, Z., Shi, Y., Lin, Z., ... and Sun, X., Incentivizing Reasoning for Advanced Instruction-Following of Large Language Models, *Advances in Neural Information Processing Systems (NeurIPS)*. 2025
- Cui, X., Qin, Y.*, Gao, Y., Zhang, E., Xu, Z., Wu, T., ... and Li, H., Sinkhorn Distance Minimization for Knowledge Distillation, *Proceedings of the Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING)*. 2024
- Qin, Y., Chen, X., Shen, Y., Fu, C., Gu, Y., Li, K., ... and Ji, R., CAPro: Webly Supervised Learning with Cross-modality Aligned Prototypes, *Advances in Neural Information Processing Systems (NeurIPS)*. 2023
- Qin, Y., Chen, X., Chen, C., Shen, Y., Ren, B., Gu, Y., ... and Shen, C., FoPro: Few-shot Guided Robust Webly-supervised Prototypical Learning, *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*. 2023
- Qin, Y., Zheng, H., Gu, Y., Huang X., Yang, J., Wang, L., Zhu, Y.M., Learning Bronchiole-Sensitive Airway Segmentation CNNs by Feature Recalibration and Attention Distillation, *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. 2020
- Qin, Y., Gu, Y., Zheng, H., Chen, M., Yang, J., Zhu, Y.M., AirwayNet-SE: A Simple-Yet-Effective Approach to Improve Airway Segmentation Using Context Scale Fusion, *IEEE International Symposium on Biomedical Imaging (ISBI)*. 2020
- Qin, Y., Chen, M., Zheng, H., Gu, Y., Shen, M., Yang, J., Huang, X., Zhu, Y.M., Yang, G.Z., AirwayNet: A Voxel-Connectivity Aware Approach for Accurate Airway Segmentation Using Convolutional Neural Networks, *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. 2019
- Qin, Y., Zheng, H., Zhu, Y.M., Yang, J., Simultaneous Accurate Detection of Pulmonary Nodules and False Positive Reduction Using 3D CNNs, *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2018

SERVICES

Area Chair

International Conference on MICCAI 2025,2026

Journal Reviewer

I regularly review for IEEE TMI, IEEE TNNLS, IEEE JBHI, MedPhys, AIM, CMPB, MLA, and IJCARS.

Conference Reviewer

I regularly review for MICCAI, IPCAI, ACLs, AAAI, NeurIPS, CVPR, ICCV, ICLR, and ICML.

SKILLS

Languages: Mandarin (native), English (TOELF 105), Japanese (JLPT N1), French (un peu)